No class on 11/3 due to elections

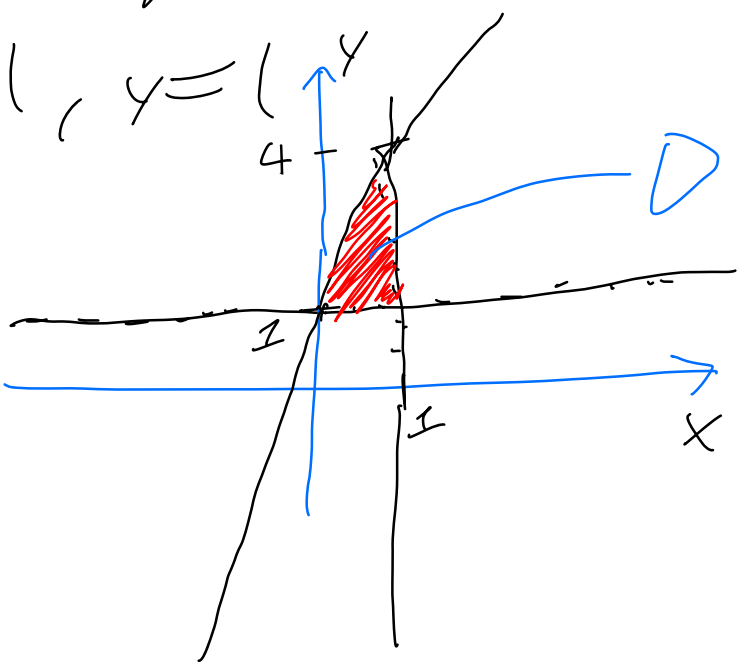
Poll Everywhere starts this week. Was supposed to start last week, just that nobody told me. Will be at the end of each discussion.

Poll Everywhere link: pollev.com/efn1738, there will also be a QR code later

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1: $\iint_D f\,dA$, D bounded by $y = 3x + 1$, $x = 1$, $y = 1$

2:



b, c: dydx $\Rightarrow$ x on outside $\Rightarrow$ x bounds

are constant $\Rightarrow 0 \le x \le 1$. Highest y can go up to is $3x+1$, so $1 \le y \le 3x+1$, and it's

$$\int_0^1 \int_1^{3x+1} f \, dy \, dx.$$

$dx\,dy \Rightarrow y$ outside $\Rightarrow y$ c.b. $\Rightarrow 1 \le y \le 4$.

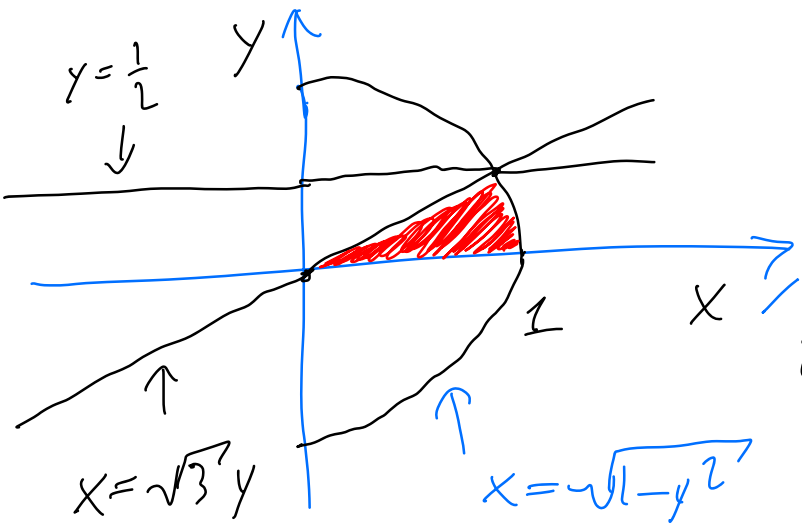
Lowest value of x is on diagonal line

$y = 3x+1 \Rightarrow x = \frac{y-1}{3} \Rightarrow \frac{y-1}{3} \le x \le 1$, so it's

$$\int_1^4 \int_{\frac{y-1}{3}}^1 f \, dx \, dy.$$

5: Sketch, polarize, find

$$\int_0^{1/2} \int_{\sqrt{3}y}^{\sqrt{1-y^2}} xy^2 \, dx \, dy$$



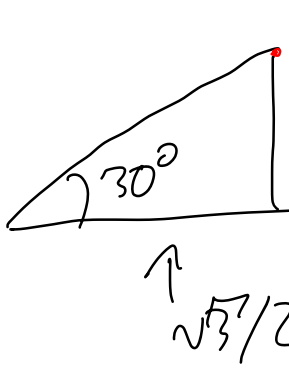
$x = \sqrt{1-y^2} \Rightarrow x^2 = 1-y^2$

$\Rightarrow x^2+y^2=1$, so this is a $\frac{1}{2}$-circle.

Notice that all curves intersect at $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

because that point satisfies all equations.

Note this is a sector of unit circle, so $0 \le r \le 1$. From diagram, we know that $0 \le \theta \le \frac{\pi}{6}$. Now plug in $dx\,dy = r\,dr\,d\theta$, and also $xy^2 = r^3 \cos\theta \sin^2\theta$, so it's



$$\int_0^1 \int_0^{\pi/6} r^4 \cos\theta \sin^2\theta \, d\theta\, dr = \underbrace{\left(\int_0^1 r^4\, dr\right)}_{A} \underbrace{\left(\int_0^{\pi/6} \cos^2 d\theta\right)}_{B}$$

$$A = \frac{r^5}{5}\Big|_0^1 = \frac{1}{5}$$

$$B = \int_0^{\pi/6} \cos \cdot \sin^2 = \frac{\sin^3\theta}{3}\Big|_0^{\pi/6} = \frac{\sin^3\left(\frac{\pi}{6}\right)}{3} = \frac{1}{24},$$

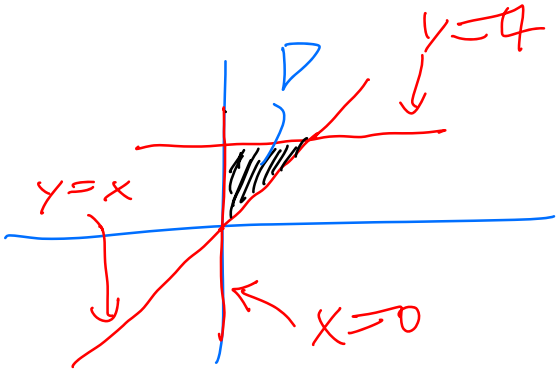
so $AB = \frac{1}{5} \cdot \frac{1}{24} = \boxed{\frac{1}{120}}$.

2: Average of $f = y^2 e^{xy}$ over D bounded by

$y=x$, $y=4$, $x=0$.

Recall: $\text{avg}_D f = \dfrac{\iint_D f\,dA}{\text{area}(D)}$.

To find area, we have to graph D.



Right triangle with legs of length 4, so area $= \frac{1}{2}\cdot 4^2 = 8$.

Notice y^2e^{xy} is easier to integrate with respect to x because 1 factor of y will just vanish. So we want to put y on the outside. So $0\le y\le 4$ & $0\le x\le y$.

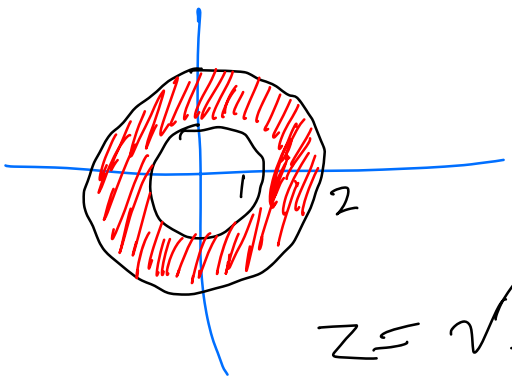
$$\iint f\,dA = \int_0^4 \left(\int_0^y y^2 e^{xy}\,dx\right) dy = \int_0^4 ye^{xy}\Big|_{x=0}^{x=y}$$

$$= \int_0^4 \left(ye^{y^2} - y\right) dy = \frac{1}{2}e^{y^2} - \frac{y^2}{2}\Big|_0^4 =$$

$$\frac{1}{2}e^{16} - 8 - \frac{1}{2} = \frac{e^{16}-17}{2}, \text{ so average} =$$

$$\iint \text{area} = \boxed{\frac{e^{16}-17}{16}}.$$

4: Volume below $z=\sqrt{x^2+y^2}$ above $1\le x^2+y^2\le 4$.



Recall: $x=r\cos\theta$, $y=r\sin\theta$, $x^2+y^2=r^2$, $dA=dx\,dy=r\,dr\,d\theta$, so

$z=\sqrt{x^2+y^2}=\sqrt{r^2}=r$ since $r\ge 0$.

Annulus $\Rightarrow$ no restrictions on θ, so $0\le\theta\le 2\pi$, and $1\le r^2\le 4 \Rightarrow 1\le r\le 2$.

z is going from 0 to r, so volume is

$$\iint_A r\,dA = \int_0^{2\pi}\int_1^2 r^2\,dr\,d\theta =$$

$$\left(\int_0^{2\pi} d\theta\right)\left(\int_1^2 r^2\,dr\right) = 2\pi\cdot\frac{r^3}{3}\Big|_1^2 = 2\pi\cdot\frac{7}{3} = \boxed{\frac{14\pi}{3}}.$$

7a: $\iiint_E (xy+z^2)\,dV$, $E=[0,2]\times[0,1]\times[0,3]$.

From E's description, $0\le x\le 2$, $0\le y\le 1$, $0\le z\le 3$, so

$$\text{int} = \int_0^2\int_0^1\int_0^3 (xy+z^2)\,dz\,dy\,dx =$$

$$\int_0^2 \int_0^1 \left(xyz + \frac{z^3}{3}\right)\Big|_{z=0}^{z=3} = \int_0^2 \int_0^1 (3xy + 9)\, dy\, dx$$

$$= \int_0^2 \left(1.5xy^2 + 9y\right)\Big|_{y=0}^{y=1} = \int_0^2 (1.5x + 9)\, dx =$$

$$0.75x^2 + 9x\Big|_0^2 = 3 + 18 = \boxed{21.}$$

7b: $\iiint_E \frac{1}{x^3}\, dV$, $E = \{0 \le y \le 1,\ 0 \le z \le y^2,\ 1 \le x \le z+1\}$

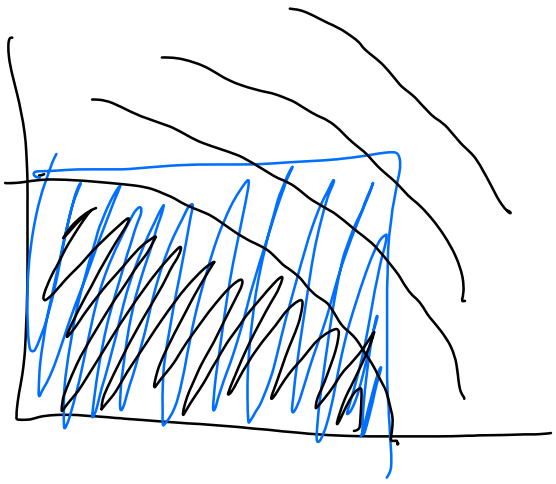
1st step is to convert description of E into region bounds. So it's $\int_0^1 \int_0^{y^2} \int_1^{z+1} \frac{1}{x^3}\, dx\, dz\, dy$

$$= \int_0^1 \int_0^{y^2} -\frac{1}{2x^2}\Big|_{x=1}^{x=z+1} = \int_0^1 \int_0^{y^2} \frac{1}{2}\left(1 - \frac{1}{(z+1)^2}\right) dz\, dy$$

$$= \int_0^1 \frac{1}{2}\left(z + \frac{1}{z+1}\right)\Big|_{z=0}^{z=y^2} = \frac{1}{2}\int_0^1 \left(y^2 + \frac{1}{y^2+1} - 1\right) dy$$

$$= \frac{1}{2}\left(\frac{1}{3}y^3 + \arctan^{-1} y - y\right)\Big|_0^1 = \frac{1}{2}\left(\frac{1}{3} + \frac{\pi}{4} - 1\right) =$$

$$\frac{1}{2}\left(\frac{\pi}{4} - \frac{2}{3}\right) = \boxed{\frac{3\pi - 8}{24}}.$$

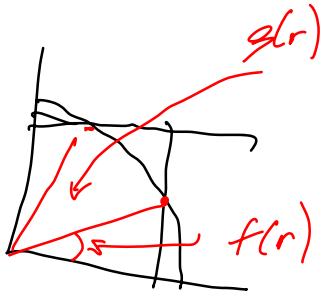


$0 \le x \le 1$
$0 \le y \le 1$ → r & θ bounds

$0 \le r\cos\theta \le 1$
$0 \le r\sin\theta \le 1$

$0 \le r \le 1$
$0 \le \theta \le \frac{\pi}{2}$

$1 \le r \le \sqrt{2}$
$r(r) \le \theta \le g(r)$



$$\int_0^1\int_0^1 h\,dx\,dy = \int_0^1\int_0^{\frac{\pi}{2}} f(\cdots)\,r\,dr\,d\theta + \int_1^{\sqrt{2}}\int_0^{g(r)} h\,r\,dr\,d\theta$$

6: Sketch, polarize, find

$$\int_0^1\int_0^1 + \int_1^{\sqrt{2}}\int_0^x + \int_{\sqrt{2}}^2\int_0^{\sqrt{4-x^2}} xy\,dy\,dx$$

$$\int_{1/\sqrt{2}}^{1}\int_{\sqrt{1-x^2}}^{x} + \int_1^{\sqrt{2}}\int_0^x + \int_{\sqrt{2}}^2\int_0^{\sqrt{4-x^2}}$$



$y = \sqrt{4-x^2} \Rightarrow x^2+y^2=4$
⇒ circle radius 2

$\frac{1}{\sqrt{2}}$ 1 $\sqrt{2}$ 2

Notice: D is a sector between circles of radius 1 & 2, so $1 \le r \le 2$. $y = x$ is at a 45° angle to the x-axis & $45° = \frac{\pi}{4}$, so $0 \le \theta \le \frac{\pi}{4}$.

Notice $xy\,dy\,dx = (r^2 \sin\theta \cos\theta)(r\,dr\,d\theta) = r^3 \sin\theta\cos\theta\,dr\,d\theta$,

so A's $\int_1^2 \int_0^{\pi/4} r^3 \sin\theta\cos\theta\,d\theta\,dr = \left(\int_1^2 r^3\,dr\right)\left(\int_0^{\pi/4} \frac{1}{2}\sin 2\theta\,d\theta\right)$

A B

$A = \frac{r^4}{4}\Big|_1^2 = \frac{15}{4}$

$B = -\frac{1}{4}\cos 2\theta\Big|_0^{\pi/4} = \frac{1}{4}$, so $AB = \boxed{\frac{15}{16}}$.